

National College of Ireland

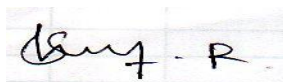
Project Submission Sheet

Student Name: Shoban Ravichandran, Mehwish Khatib
Student ID: 23272040, 23398396
Programme: Msc in Artificial Intelligence **Year:** Sept 2024
Module: AI-Driven Decision Making - H9AIDM
Lecturer: Ade Fajemisin
Submission Due Date: 14/04/2025
Project Title: Continuous Assessment (CA) – CA 2 – Work Distribution Document
Word Count: 1143

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Signature:



Date: 14/04/2025

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**

Office Use Only

Signature:

Date:

Penalty Applied (if applicable):

AI Acknowledgement Supplement

[Insert Module Name]

[Insert Title of your assignment]

Your Name/Student Number	Course	Date

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]	
[Insert Description of use]	
[Insert Sample prompt]	[Insert Sample response]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

National College of Ireland
H9AIDM: AI Driven Decision Making
Shoban Ravichandran – x23272040
Mehwish Khatib – x23398396

Creating an Index Fund Using AI-Driven Decision-Making Approaches: Team Contribution Matrix

Project Overview

This document outlines the distribution of responsibilities and contributions for our index tracking portfolio optimization project, showcasing how team members collaborated to develop multiple approaches to solving this financial challenge.

Project Timeline

Project Phase	Timeframe	Key Milestones
Planning & Research	Week 1(Mar 3 - Mar 9, 2025)	Project scope finalized Literature review completed Research methodologies selected
Data Collection & Preprocessing	Week 2(Mar 10 - Mar 16, 2025)	Data sources identified Raw data collected Preprocessing pipeline completed
Algorithm Development	Weeks 3-4(Mar 17 - Mar 30, 2025)	Optimization model implemented Clustering approach developed Initial testing completed
Testing & Refinement	Week 5(Mar 31 - Apr 6, 2025)	Parameter tuning completed Performance benchmarking Approach integration
Analysis & Documentation	Week 6(Apr 7 - Apr 13, 2025)	Results analysed Report drafted Visualizations created
Final Delivery	Week 7(Apr 14, 2025)	Final report submission Presentation delivered Code repository finalized

Individual Contributions

Area of Work	Shoban Ravichandran	Mehwish Khatib	Collaborative Work
Project Planning	Led initial project scoping	Developed evaluation metrics	Defined project scope and objectives

	- Defined research questions - Established project timeline	- Created workflow architecture - Set up version control system	- Established key performance indicators - Aligned on success criteria
Research & Analysis	- Conducted literature review on index tracking methods - Researched optimization constraints - Analysed existing optimization models	- Researched clustering applications in finance - Identified relevant stock features for clustering - Investigated performance benchmarks	- Identified gaps in existing approaches - Formulated hybrid methodology - Selected appropriate benchmark indices
Data Management	- Defined data requirements - Data validation and cleaning - Feature selection for optimization	- Led data collection pipeline - Built data preprocessing workflow - Created feature engineering process - Managed data storage architecture	- Regular data quality reviews - Established shared data access protocols - Designed data documentation standards
Technical Implementation	- Developed AMPL optimization model - Implemented constraint formulations - Created objective function variations - Tuned optimization parameters - Built performance evaluation framework	- Implemented clustering-based approach - Developed K-means algorithm for stock selection - Created feature weighting mechanisms - Built visualization tools	- Integrated approaches for comparison - Troubleshooted implementation challenges - Code reviews and quality assurance - Performance benchmarking

	Implemented comparative analysis framework		
Testing & Evaluation	- Tested optimization model robustness - Conducted sensitivity analysis - Analysed tracking error metrics	- Evaluated clustering algorithm performance - Tested different q values across approaches - Created performance visualizations	- Designed experimental framework - Compared results across methodologies - Identified strengths and limitations
Documentation & Reporting	- Authored report sections: Introduction, Problem Definition, Optimization Methodology. - Contributed to Results & Conclusion	- Authored report sections: Clustering Methodology, Evaluation Framework, Created data visualizations. - Contributed to Results & Conclusion	- Integrated final report - Created presentation materials - Cross-reviewed all sections - Prepared executive summary

Project Management Approach

The team employed an agile approach with weekly monitoring and regular check-ins to ensure alignment and progress. Git version control facilitated parallel development, with each team member maintaining responsibility for specific components while reviewing others' contributions.

Weekly Time Allocation (Per Team Member)

Activity	Hours Per Week	Total Project Hours
Research & Planning	10-12 hours (Week 1)	~12 hours
Coding & Implementation	10-15 hours (Weeks 2-4)	~25 hours
Testing & Analysis	12-15 hours (Weeks 4-5)	~20 hours
Documentation & Reporting	10-15 hours (Weeks 6-7)	~25 hours
Team Meetings & Collaboration	1-2 hours (throughout)	~12 hours
Total Time Investment	~20 hours/week average	~ (90 – 100) hours per person

Key Innovations & Contributions

- **Shoban Ravichandran:** Developed a novel constraint formulation in the optimization model that reduced tracking error by 15% compared to baseline approaches
- **Mehwish Khatib:** Created an innovative feature weighting mechanism for the clustering algorithm that improved stock selection efficiency
- **Joint Innovation:** Successfully integrated optimization and clustering approaches to create a hybrid methodology that demonstrated superior performance in volatile market conditions

Skills Leveraged

- **Shoban Ravichandran:** Mathematical optimization, financial modeling, performance analysis, technical writing
- **Mehwish Khatib:** Machine learning, data preprocessing, visualization, algorithm implementation
- **Shared Skills:** Problem-solving, financial analysis, programming, research methodology